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Top Skills

AI Governance
Retrieval-Augmented Generation (RAG)
AI Solutions Architecture

Languages

Nepali (Native or Bilingual)
Hindi (Native or Bilingual)
English (Native or Bilingual)

Certifications

SACHe
Productivity and Time Management for the Overwhelmed
Project Management Professional (PMP)®
Honored Listee

Abhishek Shah

AI Solutions Architect | Production Multi Agent Systems and Enterprise AI Governance | Industrial and AEC
Beaverton, Oregon, United States

Summary

I am an AI solutions architect at SSOE Group, an architecture, engineering, and construction firm with 1,400 employees. I was the main technical architect for Pathfinder, a production system in which multiple AI agents search the firm's documents and draft answers for business development and proposal work. More than 200 executives use it. I also led the firm wide launch of the Abacus AI platform, which reached 80 percent adoption in 30 days. Employees have built about 900 apps on the platform, and I built the process the firm uses to review and publish them. Before SSOE, I built AI systems for automotive and semiconductor manufacturers. At BMW of North America, I shipped a pipeline on AWS Bedrock that 25 attorneys use to process contracts. I also ran the AI Champions program across BMW plants in the United States, Germany, and China. At Intel, I spent five years as a project engineer, data scientist, and AI program manager in semiconductor fabs. At Intel Labs, I built computer vision models that run on edge devices. I started my career as a chemical process engineer at Westlake Chemical. That background lets me talk with plant engineers and operators about their own processes, and it shapes which problems I think AI should solve. I hold a master's degree in AI and machine learning from the University of Michigan, earned with distinction, and a PMP certification. My recent open source projects predict part quality on Bosch production line data, detect defects with vision models on edge hardware, triage wafer defects, and serve predictive maintenance models on AWS. I am looking for AI architect and AI program leadership roles at industrial, manufacturing, and AEC companies.

Experience

SSOE Group

AI Solution Architect and Technical Program Manager
September 2025 - Present (1 year 2 months)

- Architected a production multi agent retrieval augmented generation system for business development and proposal preparation, with routing by intent, hybrid retrieval that combines BM25 keyword search with dense vector search, reconciliation of evidence across sources, and access control through Microsoft Entra ID; all 200+ executives with access now use it.
- Launched Abacus AI, the firm's agentic AI platform, to 1,400 employees through a program covering governance, data architecture, security, adoption, and developer enablement; reached 80 percent adoption in 30 days and full deployment by day 90, then managed the platform ontology, token budgets, and dashboards that monitor usage and cost.
- Enabled citizen developers to build about 900 agentic AI apps on Abacus AI, and built the SSOE AI App Store on SharePoint to distribute them, with private and public catalogs, a three stage review chain that ends at an admin publish gate, and scoring criteria that decide which apps the firm supports.
- Delivered two retrieval augmented generation products from requirements through vendor build, validation, and production handoff: an AI assistant for engineers that reached 100 percent adoption in that group, and an estimator that drafts early project costs from past project documents and cut estimate delivery time by 20 percent.
- Defined how teams build, test, secure, and publish agents in Microsoft Copilot Studio, decided which workloads run on Copilot and which run on Abacus AI, and built an adoption framework with IT security that routes each new AI capability through security review, change management, and a time boxed Entra ID pilot.

BMW of North America

AI Product Lead / Data Scientist

October 2024 - September 2025 (1 year)

- Shipped an agentic retrieval augmented generation pipeline on AWS Bedrock, with Claude 3.5 Sonnet as the model, that 25 attorneys across North America use to process contracts; the pipeline cut the cycle time for processing contracts by 15 percent and saved each legal team 10 hours of work per week.
- Built a containerized FastAPI inference microservice in Python that detects symbols in technical manuals, using asyncio to run vision model and LLM calls in parallel and Pydantic to define and validate the contract for every request and response the service handles.
- Delivered an XGBoost model through DataRobot that detects paint defects with 92 percent accuracy and cut scrap cost by \$70K per month, and built

separate forecasting models that predict inventory needs and transportation demand for BMW of North America operations.

- Ran the BMW AI Champions program across the Spartanburg, Germany, and China sites, training plant teams to identify and run manufacturing AI use cases on their own, and trained about 150 employees each week on BMW's internal Group AI Assistant.

Intel Corporation

5 years

Project Engineer, Data Scientist, and AI Program Manager

November 2019 - October 2024 (5 years)

Hillsboro, OR

- Built and deployed a retrieval augmented generation pipeline that pairs the OpenAI API with a vector database for 45 engineers, eliminating 10 hours of manual work per engineer each week and producing an estimated \$100K per month in soft savings for the engineering organization.
- Built machine learning and statistical process control models on tool sensor and process data in high volume fab manufacturing, flagging process excursions earlier and giving engineers faster root cause signals, after joining Intel as a fab project engineer who managed process and equipment projects.
- Built the telemetry and measurement layer with Airflow, PySpark, Azure Data Factory, and Power BI, replacing manual reporting with automated data pipelines and dashboards that let engineers track throughput, failure modes, and bottlenecks and confirm whether each improvement held over time.

Data Engineer / Project Manager

February 2022 - June 2022 (5 months)

Hillsboro, Oregon, United States

Created Data validation dashboard by using Azure DF analytics for 100+ DF pipelines for ML,

PowerBI by using stream analytics and designed concepts for the Dashboard, UAT, deployment, and developing functionality.

Created a chatbot using the Azure cognitive language service for Q&A to improve the overall interaction within our organization and helped reduce the information request tickets by 30% which improved team's productivity.

Intel Labs

Machine Learning Engineer

June 2022 - April 2023 (11 months)

Hillsboro, Oregon, United States

AI Edge computing Machine Learning Engineer

Westlake Chemical

Data Analytics Process Engineer

September 2017 - November 2019 (2 years 3 months)

Portland, Oregon Area

Utilized data analytics and statistical process control to identify critical pathways and design effective safeguards for a new unit operation for recovering brine from waste, resulting in a 20% reduction in overall raw material usage.

Conducted a conceptual design study utilizing data and analytics to evaluate the feasibility of a new \$4MM chlorine liquefaction system, leading to the selection of an environmentally friendly alternative that increased system throughput by 15% and improved safety performance.

Applied data analytics and predictive modeling to redesign an acid discharge pump design, resulting in a 25% reduction in acid consumption and an increase in process efficiency.

Developed a data-driven safe work procedure for collecting stack samples from HCl burners and trained internal lab personnel, leading to an annual cost savings of \$20,000 in outside lab consultation services.

Served as a process safety management liaison and hazard analysis and operability study facilitator, using big data and analytics to review plant process changes for potential health, safety, and environmental impacts.

Managed three construction and commissioning projects during a plant shutdown, supervising a team of 10 contractors and using data analytics to optimize project schedules and resources, resulting in a 5% reduction in project timelines.

Tate & Lyle

5 years 4 months

Staff Data Analytics Process Engineer & Project Manager

September 2016 - August 2017 (1 year)

Duluth, Minnesota Area

Data Analytics Process Engineer

May 2012 - September 2016 (4 years 5 months)

Duluth, Minnesota Area

Managed a total of \$5MM+ capital projects by using data-driven decision-making to provide research for both engineering and construction phases and estimate capital and utility value with an AFE.

Provided technical assistance through the product lifecycle of the project, collaborating with the core team to use data-driven decision-making to perform HAZOP for the equipment installation and identify risks and add layers of protection for safe operation.

Developed and managed key process indicators (KPIs) for the plant using data analytics to track production data, raw material and ingredient usage, utility usage, yield, and % uptime on every unit operation. A total of 150 metrics were tracked on a daily basis to inform crucial decisions on process optimization, debottlenecking, production planning, and downtime reduction.

Used data analytics to identify trends and patterns in the data and make recommendations for improvement.

Served as PI data link administrator for the plant to maintain, control, and secure the plant's database information.

Developed a dynamic dashboard on the PI process book using data-driven decision making to allow operators to monitor plant processes in real-time, improving visibility and decreasing troubleshooting response time, and reducing soft cost.

Utilized data-driven decision making during outages as a turnaround engineer, collaborating with the production and maintenance department to schedule, train, and supervise contractors. Conducted JSAs, reviewed and verified lockouts, and completed hazard mitigation plans using data-driven decision-making to identify potential risks and hazards.

Worked on 10 outages to date while supervising up to 20 contractors, ordering and sizing equipment and managing outage budgets using analytics to optimize resource allocation and minimize costs, and assisting in plant start-up.

Education

University of Michigan

Master of Science - MS, Artificial Intelligence and Machine Learning

University of Minnesota

Bachelor's degree, Chemical Engineering

Washington State University

Master of Engineering and Technology Management METM, Information
Technology